

# Uttkarsh Thakur

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## SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, SQL, C++

**AI/ML:** RAG Pipelines, Semantic Search, Vector Databases, FAISS, AST Parsing, NLP, Transformers, LLMs, OpenAI API

**ML Libraries:** LangChain, Sentence-Transformers, TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas

**Backend & Web:** FastAPI, Flask, Spring Boot, Spring Data JPA, REST API Design, React.js, Node.js, HTML/CSS

**Databases:** PostgreSQL, MySQL, MongoDB, H2, Flyway Migrations

**Developer Tools:** Git & GitHub, Docker, Maven, JUnit 5, Jupyter Notebook, Google Colab, VS Code

## PROJECTS

**CodeNav AI** | *FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython* 2026

- Built an **AI-powered search tool** that lets a developer ask questions about an unfamiliar codebase in plain English and get back the relevant code, using **large language models** and **semantic search (RAG)**.
- Improved answer quality by splitting code along real function and class boundaries instead of fixed-size blocks, so results come back complete and in context.
- Cut the time to open a codebase from **over 20 seconds to under 0.1 seconds**, a **99.5% improvement**, by saving the search index to disk instead of rebuilding it each time.
- Delivered the tool as a web application with a **React.js** interface and a **FastAPI** backend.

**BillSync** | *Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker* 2026

- Built a **full-stack web application** for splitting group expenses: members log what they paid, and the app works out who owes whom, backed by a **Java Spring Boot** REST API of **16 endpoints** and a **React** and **TypeScript** interface.
- Solved a rounding problem so every split adds up to the **exact total** with no lost fractions, and developed a **settle-up feature** that turns a tangle of debts into the fewest payments possible.
- Designed the **PostgreSQL** database with version-controlled migrations and wrote **73 automated tests**; tuned a slow screen that made **22 database calls** down to **1**.

**Motion Sensing System** | *TensorFlow/Keras, VGG16, Flask, Streamlit, OpenCV, Python* 2024

- Built a system that **recognises human activities from live video**, adapting a pre-trained image-recognition model (**VGG16**) rather than training one from scratch.
- Delivered it as a live dashboard that displays activity predictions as they happen.

## RESEARCH & TALKS

**Agreement Loops in Multi-Agent Systems** | *Seminar Talk, Literature Review* 2026

- Reviewed current research on **how independent AI agents reach agreement** with one another across the wider field of distributed AI.
- Presented the findings as a **seminar talk**, covering what works reliably today and which problems remain unsolved.

## EDUCATION

**Bennett University** Aug. 2023 – May 2027

*B.Tech in Computer Science and Engineering* | **CGPA: 8.17/10** Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

**DAV Public School** Graduated 2022

*Senior Secondary Education (Class XII)* Ranchi, India

## CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera Nov. 2024

## ACHIEVEMENTS

Ranked **101<sup>st</sup> nationally** at **Smart India Hackathon 2024**

**CodeChef:** 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)